
2. What are semantic HTML tags?

Answer:

Semantic tags clearly describe their meaning to both developers and browsers. Examples include `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>`. They improve readability, accessibility, and SEO.

3. Difference between id and class?

Answer:

An id is unique and should be used for a single element on a page. A class can be used for multiple elements that share the same styling or behavior.

4. What is Flexbox?

Answer:

Flexbox is a CSS layout model used to align and distribute items efficiently within a container. It helps create responsive layouts with less code.

5. What is CSS Grid?

Answer:

CSS Grid is a two-dimensional layout system that allows developers to arrange elements into rows and columns for complex webpage layouts.

6. Difference between relative, absolute, and fixed positioning?

Answer:

- Relative: Positioned relative to its normal position.
 - Absolute: Positioned relative to its nearest positioned parent.
 - Fixed: Positioned relative to the browser window and remains fixed during scrolling.
-

7. What is Responsive Web Design?

Answer:

Responsive Web Design ensures a website looks good on all devices such as

mobiles, tablets, and desktops by using flexible layouts, media queries, and responsive units.

8. How do you center a div?

Answer:

I usually use Flexbox:

```
display: flex;  
justify-content: center;  
align-items: center;
```

This centers the div both horizontally and vertically.

JavaScript

9. Difference between var, let, and const?

Answer:

- var is function-scoped and can be redeclared.
 - let is block-scoped and can be reassigned.
 - const is block-scoped and cannot be reassigned after initialization.
-

10. What are closures?

Answer:

A closure is a function that remembers and can access variables from its outer scope even after the outer function has finished executing.

11. What is hoisting?

Answer:

Hoisting is JavaScript's behavior of moving variable and function declarations to the top of their scope before execution.

12. Difference between == and ===?

Answer:

- `==` compares values after type conversion.
- `===` compares both value and data type without type conversion.

Example:

```
5 == "5" // true
5 === "5" // false
```

13. What are Promises?

Answer:

Promises are used to handle asynchronous operations. They can be in three states: pending, fulfilled, or rejected.

14. What is `async/await`?

Answer:

`Async/await` is a cleaner way to work with Promises. It makes asynchronous code look synchronous and improves readability.

15. What is the DOM?

Answer:

DOM stands for Document Object Model. It represents the HTML document as a tree structure and allows JavaScript to modify webpage content dynamically.

16. What is event bubbling?

Answer:

Event bubbling means an event starts from the target element and propagates upward through its parent elements.

17. What are arrow functions?

Answer:

Arrow functions are a shorter syntax for writing functions in JavaScript and do not have their own `this` binding.

Example:

```
const add = (a,b) => a+b;
```

18. Difference between map(), filter(), and reduce()?

Answer:

- map() transforms each element.
 - filter() returns elements that satisfy a condition.
 - reduce() combines all elements into a single value.
-

React

19. What is React?

Answer:

React is a JavaScript library used for building fast and interactive user interfaces using reusable components.

20. What is Virtual DOM?

Answer:

Virtual DOM is a lightweight copy of the real DOM. React updates the Virtual DOM first and then updates only the changed parts in the real DOM, improving performance.

21. Difference between props and state?

Answer:

- Props are data passed from parent to child components.
 - State is data managed within a component and can change over time.
-

22. What are React Hooks?

Answer:

Hooks are functions that allow us to use state and lifecycle features in functional components.

23. What is useState?

Answer:

useState is a React Hook used to create and manage state in functional components.

24. What is useEffect?

Answer:

useEffect is a React Hook used for handling side effects such as API calls, timers, and event listeners.

25. Why do we use keys in React lists?

Answer:

Keys help React identify which items have changed, been added, or removed, making rendering more efficient.

26. What is Redux?

Answer:

Redux is a state management library used to manage and share application state across multiple components.

27. What is React Router?

Answer:

React Router is a library used for navigation between different pages in a React application without refreshing the page.

28. What is JSX?

Answer:

JSX stands for JavaScript XML. It allows us to write HTML-like code inside JavaScript.

29. What is lifting state up?

Answer:

Lifting state up means moving state to a common parent component so multiple child components can access and share the same data.

Node.js**30. What is Node.js?****Answer:**

Node.js is a JavaScript runtime environment that allows JavaScript code to run outside the browser.

31. Why is Node.js single-threaded?**Answer:**

Node.js uses a single-threaded event loop to handle multiple requests efficiently without creating multiple threads.

32. What is npm?**Answer:**

npm stands for Node Package Manager. It is used to install and manage packages and dependencies.

33. What is package.json?**Answer:**

package.json is a file that contains project information, dependencies, scripts, and metadata.

34. What is middleware?**Answer:**

Middleware is a function that executes between receiving a request and sending a response. It can modify requests, responses, or perform validations.

35. Difference between synchronous and asynchronous programming?

Answer:

- Synchronous code executes one task at a time.
 - Asynchronous code allows other tasks to continue while waiting for an operation to complete.
-

36. What is the Event Loop?

Answer:

The Event Loop continuously checks the call stack and callback queue to execute asynchronous operations efficiently.

Express.js

37. What is Express.js?

Answer:

Express.js is a lightweight Node.js framework used to build web applications and REST APIs.

38. What is Routing?

Answer:

Routing determines how an application responds to client requests for specific URLs and HTTP methods.

39. Difference between GET and POST?

Answer:

- GET retrieves data from the server.
 - POST sends data to the server for processing or storage.
-

40. What is REST API?

Answer:

REST API is an architectural style that uses HTTP methods like GET, POST, PUT, and DELETE to perform CRUD operations.

41. How do you handle errors in Express?

Answer:

I use error-handling middleware with four parameters:

```
app.use((err, req, res, next) => {  
  res.status(500).json({message: err.message});  
});
```

42. What is CORS?

Answer:

CORS (Cross-Origin Resource Sharing) allows a frontend application to access resources from a different domain.

MongoDB

43. What is MongoDB?

Answer:

MongoDB is a NoSQL database that stores data in flexible JSON-like documents.

44. Difference between SQL and MongoDB?

Answer:

- SQL stores data in tables and rows.
 - MongoDB stores data in collections and documents.
-

45. What is a Document?

Answer:

A document is a single record in MongoDB stored in BSON format.

46. What is a Collection?

Answer:

A collection is a group of related documents in MongoDB.

47. What is Mongoose?

Answer:

Mongoose is an ODM library that provides schema-based interaction with MongoDB in Node.js applications.

48. What are CRUD operations?

Answer:

CRUD stands for Create, Read, Update, and Delete operations performed on database records.

49. What is Indexing?

Answer:

Indexing improves query performance by allowing MongoDB to locate data faster.

JWT & Security

50. What is JWT?

Answer:

JWT (JSON Web Token) is a secure way to transmit user information between client and server for authentication.

51. How does JWT Authentication work?

Answer:

After login, the server generates a token and sends it to the client. The client stores the token and sends it with future requests. The server verifies the token before providing access.

52. What is bcrypt?

Answer:

bcrypt is a library used to hash passwords before storing them in the database.

53. Difference between Authentication and Authorization?

Answer:

- Authentication verifies who the user is.
 - Authorization determines what the user is allowed to access.
-

54. Why should passwords be hashed?

Answer:

Hashing protects passwords from being exposed even if the database is compromised.

57. Why do you want this internship?

Answer:

"I want to gain practical industry experience, improve my technical skills, and work on real-world projects while learning from experienced developers."